

HS-1 Kinematic Suit & Robotic Teleoperation Platform — Comprehensive Technical Specification

This document presents the complete system architecture, hardware specifications, firmware protocols, 3D visualization setup, and strategic expansion roadmap for the **HS-1 Kinematic Motion Tracking Suit** and its teleoperation integration with external robotic platforms (such as the DeadGrip X-01 prototype).

1. System Architecture & Hardware Specification

1.1 Bill of Materials (BOM)

Component	Quantity	Technical Description / Application
ESP32-C3 SuperMini	14 pcs	Main processing microcontrollers (13 limb sensor nodes + 1 Master Multiplexer Hub).
Adafruit / GY BNO-055	13 pcs	9-DOF IMU absolute orientation fusion sensors.
Silicone Stranded Wire (20 AWG)	20 meters	Main power backbone (5V VBUS / 3.3V) and shared ground (GND) lines.
Perfboards (5x7 cm)	14 pcs	Prototyping circuit boards for modular node assembly.

Component	Quantity	Technical Description / Application
Pull-up Resistors	10 kOhm	Optional line stabilization for I2C communication.
Auxiliary Connectors & Passive Elements	Various	Header pins, 100nF decoupling capacitors, and textile mounts.

1.2 Internal Node Pinout Matrix (I2C Protocol)

Each individual limb node pairs a local ESP32-C3 SuperMini with a dedicated BNO-055 sensor over I2C:

Signal Line	ESP32-C3 Pin	BNO-055 Pin	Configuration / Hardware Requirement
SDA	GPIO 3	SDA / ATX	I2C Serial Data Line
SCL	GPIO 2	SCL / LRX	I2C Serial Clock Line (100 kHz)
Power (3.3V)	3V3	VIN / 3.3V	Regulated 3.3V Power Line
Ground	GND	GND	Common Shared Ground
Protocol Lock 0	—	PS0 $\rightarrow$ GND	Hardwires BNO-055 into I2C Mode
Protocol Lock 1	—	PS1 $\rightarrow$	Hardwires BNO-055

Signal Line	ESP32-C3 Pin	BNO-055 Pin	Configuration / Hardware Requirement
		GND	into I2C Mode

1.3 Inter-Module Cascaded UART Bus

Data packets propagate sequentially along limb channels via hardware UART to prevent bus contention and simplify wiring:

- **Hand Node (Node 3):** Transmits packet via GPIO 4 (Tx) $\rightarrow$ GPIO 1 (Rx) of Forearm Node.
- **Forearm Node (Node 2):** Ingests Hand packet, appends local sensor data, and transmits via GPIO 4 (Tx) $\rightarrow$ GPIO 1 (Rx) of Shoulder Node.
- **Shoulder Master Node (Node 1):** Ingests Forearm/Hand stream, appends Shoulder data, and routes stream via GPIO 4 (Tx) $\rightarrow$ Multiplexer input on the **Main ESP32-C3 Hub**.
- **Host PC Output:** USB CDC Serial line-delimited JSON stream at 115200 Baud (~50 Hz refresh rate).

2. Firmware Data Structures & Host JSON Protocol

2.1 Memory-Aligned Binary Struct (ArmPackets.h)

```
#ifndef ARM_PACKETS_H
#define ARM_PACKETS_H

#include <stdint.h>

#pragma pack(push, 1)

// Compact 14-byte binary payload for a single IMU node
struct SensorData {
    float yaw;           // 4 bytes: Euler Yaw (0.0 to 360.0 deg)
    float pitch;         // 4 bytes: Euler Pitch (-180.0 to 180.0 deg)
    float roll;          // 4 bytes: Euler Roll (-90.0 to 90.0 deg)
    uint8_t calibSys;     // 1 byte: System calibration status (0 to 3)
    uint8_t calibGyro;    // 1 byte: Gyroscope calibration status (0 to
```

```

3)
};

// Cascaded arm segment packet
struct FullArmPacket {
    SensorData hand;          // Node 3 (Hand)
    SensorData forearm;       // Node 2 (Forearm)
    SensorData shoulder;      // Node 1 (Shoulder)
};

#pragma pack(pop)

#endif // ARM_PACKETS_H

```

2.2 Serial JSON Telemetry Output Format

```

{
  "hand": { "y": 12.5, "p": -4.2, "r": 0.1, "c_sys": 3, "c_gyro": 3 },
  "forearm": { "y": 45.0, "p": 10.2, "r": -1.5, "c_sys": 3, "c_gyro":
3 },
  "shoulder": { "y": 88.1, "p": 120.0, "r": 15.3, "c_sys": 2,
"c_gyro": 3 }
}

```

3. 3D Visualizer & CAD Modeling Guidelines

3.1 Kinematic Bone Hierarchy (Ursina Engine)

- **World Origin Grid (0, 0, 0):** Master anchor point.
- **Shoulder Entity:** Parented to World; updated by Shoulder node rotation (\$Y, P, R\$).
- **Forearm Entity:** Parented to Shoulder; offset to the Elbow Pivot Joint.
- **Hand Entity:** Parented to Forearm; offset to the Wrist Rotational Joint.

3.2 Autodesk Fusion 360 Export Rules

1. **Mesh Formats:** Export isolated body segments as .OBJ files.
2. **Pivot Coordinates (0,0,0):**
 - shoulder.obj: Origin placed directly at the center of the shoulder sphere joint.
 - forearm.obj: Origin placed directly on the elbow hinge axis.

- hand.obj: Origin placed directly at the wrist rotational joint center.
3. **Axis Orientation:** \$+Y\$ aligned downward along the bone axis; \$+Z\$ aligned forward.

4. Development Roadmap & Teleoperation Milestones

Teleoperation Vision: The HS-1 full-body suit captures live motion telemetry ($\geq 100\text{ Hz}$) to drive external robotic avatars (e.g., DeadGrip X-01) with ultra-low latency.

Phase / Milestone	Key Objectives & Hardware Expansion	Target Version	Status
Phase 1	Single-arm tracking over UART cascade + Ursina Visualizer.	v1.0.0	Completed
Phase 2	• Quaternion stream integration (\$W, X, Y, Z\$) • Assembly & integration of the Main Processing Hub • Assembly & wiring of the Second Arm to the Main Hub • BVH Animation Exporter for Blender/Maya	v1.1.0	In Progress
Phase 3	• Assembly & wiring of Leg Tracking Modules to the Main Hub • Long-Range Wireless Data	v1.5.0	Planned

Phase / Milestone	Key Objectives & Hardware Expansion	Target Version	Status
	Module (ESP-NOW / Wi-Fi) • Initial full-suit physical textile assembly & wire routing • Start of Robotic Avatar (DeadGrip X-01) teleoperation integration • Comprehensive sensor calibration benchmarking & stress tests		
Phase 4	• Final 13-Node Full-Body Suit deployment • Custom Power Distribution PCB design • Full-scale robot teleoperation control & dynamic response testing • UE5 / Unity LiveLink Plugins	v2.0.0	Planned